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VHFACIntroDefTopic

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Introduction to the VHFAC Instrument

Introduction to the VHFAC Instrument

The Very High Frequency AC (VHFAC) board provides a set of instruments for testing mixed signal VHF devices. Included are two digitized waveform capture and two arbitrary waveform generator/source instruments for testing devices ranging from wireless LAN, DVD SOC, DTV, cable modems, and other mixed signal devices. In addition to the source and capture instruments, each VHFAC channel card also implements a time and jitter measurement instrument, serial bus instrument, trigger controls, and multiple PPMUs.

The VHFAC Source provides AC waveforms for applications such as video and communications device testing. The VHFAC Capture measures waveforms using a high resolution digitizer. Typical tests that can be performed are gain accuracy, frequency response, distortion, and signal-to-noise ratio.

☑ **Note:**

The VHFAC instrument requires a DSP computer in the tester configuration.

The following topics are available:

- VHFAC Instrument Overview
- VHFAC Capture Instrument
- VHFAC Serial Bus
- VHFAC Source Instrument
- VHFAC Time Measurement Unit (TMU)

- VHFAC Trigger Control
- VHFAC PPMU
- VHFAC Software Licensing Requirements
- Using VHFAC in Concurrent Testing

For more information on VHFAC instruments, see:

- About the VHFAC Capture
- About the VHFAC Serial Bus
- About the VHFAC Source
- About the Time Measurement Unit
- About the VHFAC Trigger Control

For information on VHFAC instrument performance specifications and standards, see VHFAC Specifications.
For information on DIB design, see VHFAC DIB Design Guidelines.

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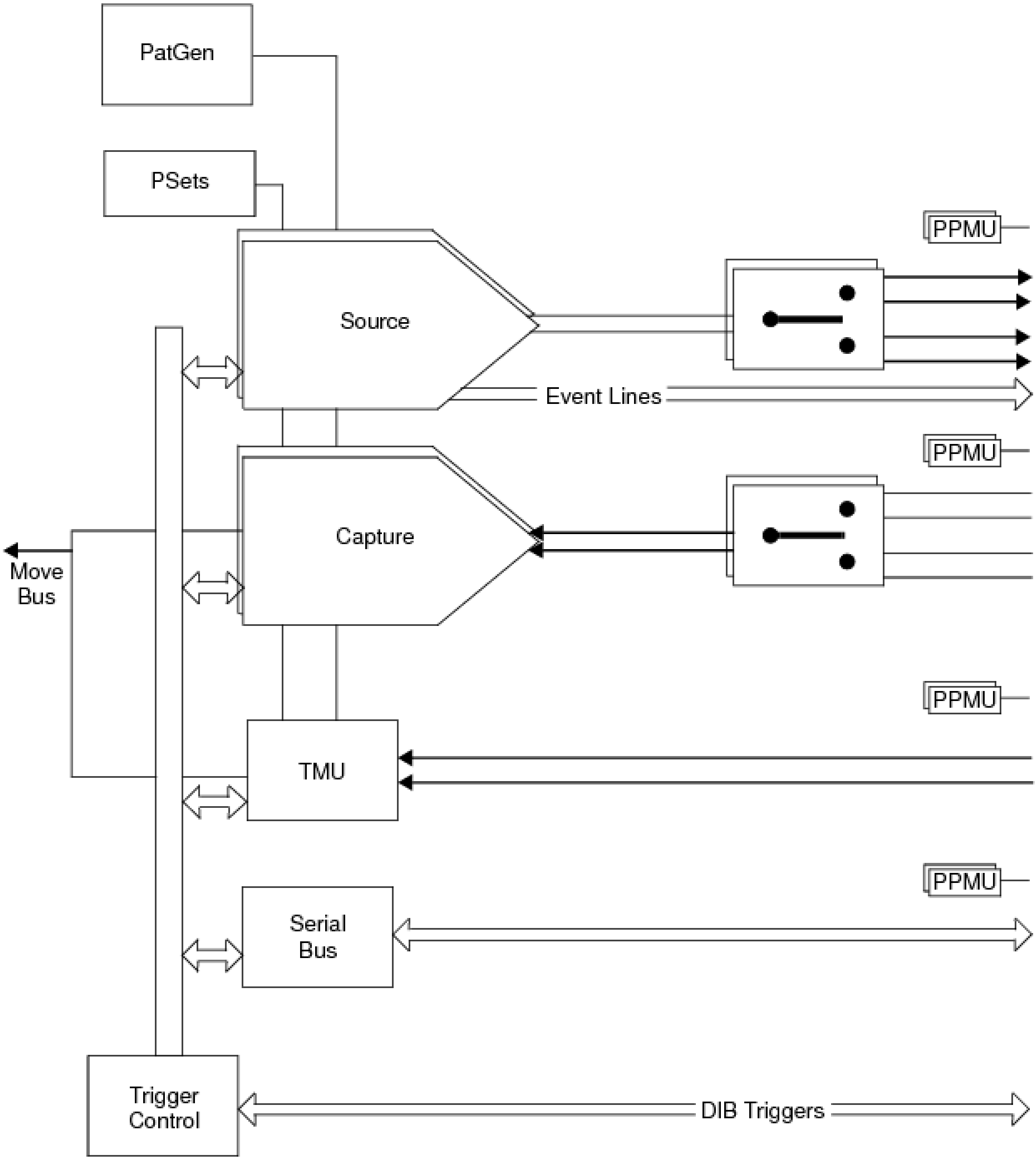
Introduction to the VHFAC Instrument : VHFAC Instrument Overview

VHFAC Instrument Overview

This figure is a block diagram of the VHFAC test option.



VHFAC Instrument

The VHFAC is composed of the following primary instruments:

- Two Source instruments per board, each with:
  - 14-bit, 400 Msps DAC
  - 8 Mega Sample Source memory

- Two Capture Instrument per board each with:
  - 14-bit at 80 Msps ADC
  - 12-bits at 125 Msps ADC
  - 1 Meg Sample capture memory
  - 14-bit, 300 MHz undersampling
  - 14-bit, peak detect
- One Time Measurement Instrument per board:
  - 20 s (low resolution), 10 ms (high resolution) time range
  - 5 ns (low resolution), 2.44 ps (high resolution) resolution
  - 80 ns (normal), 6.25 ns (interleave) time between stamps
- One Serial Bus Instrument per board, capable of I^2C and generic serial digital
- The VHFAC board also features:
  - Trigger Control (both DIB and internal instrument initiated)
  - Multiplexed single-ended or differential inputs and outputs
  - Capture Instrument initiated Move function
  - Independent instrument clocks

– PPMU per I/O

Other features of the VHFAC instrument board are:

- A total of 23 Pin Parametric Measurement Unit (PPMU) circuits, which can be used for parametric testing.
- One Pattern Generator shared by all VHFAC instruments on the board.
- Move bus connections from the VHFAC Capture instruments to the test system's support board for high speed data transfer and DSP processing.
- DIB Access connections per instrument channel
- An internal analog loopback connection between VHFAC Source and Capture instruments is available
- An external trigger connection for starting and triggering VHFAC instruments
- A set of Device Ground Sense (DGS) connections (DGS_Src_1, DGS_Src_2, DGS_Cap_1, DGS_Cap_2, TMU_DGS and SB_DGS) shared by the PPMU circuits for each instrument

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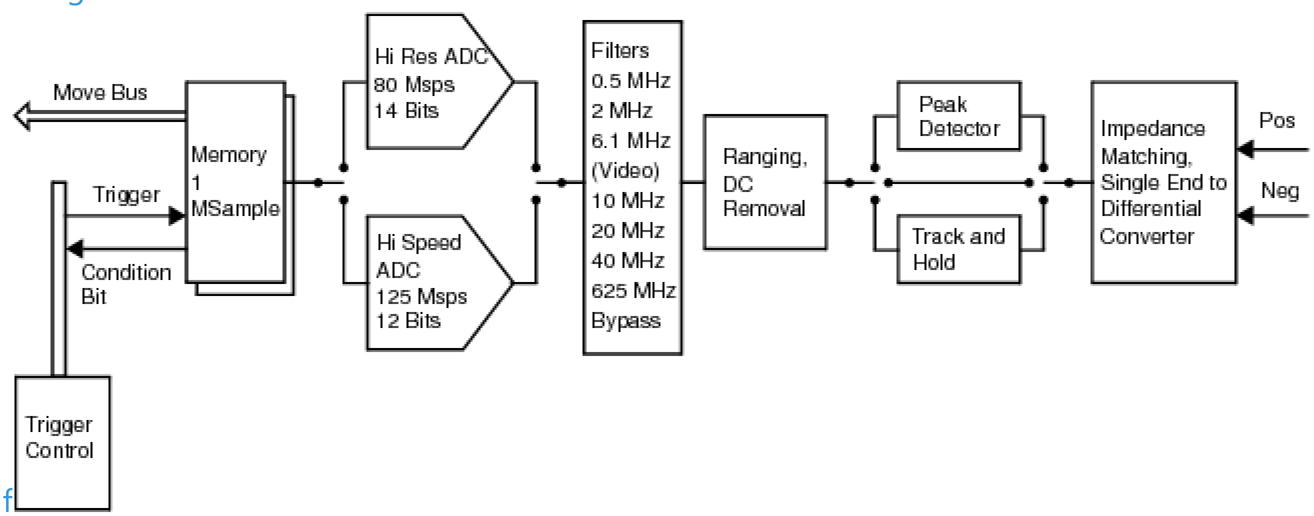
Introduction to the VHFAC Instrument : VHFAC Capture Instrument

VHFAC Capture Instrument

Each VHFAC board includes two independent capture instruments. The figure is a block diagram of a single instrument.



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VHFAC Capture Instrument

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Introduction to the VHFAC Instrument : VHFAC Capture Instrument : Features

Features

The features of the VHFAC capture instrument are:

- High speed and high resolution ADCs
- Harmonics: -60dBc @ 5MHz (128mV to 2.048V), -65dBc @ 5MHz (8mV to 128mV typical)
- Spurs: -60dBc @ 5MHz (128mV to 2.048V), -65dBm @ 5MHz (8mV to 128mV)
- Undersampling: DC to 300MHz, Max Fs = 20MHz
- Wide filter selection (0.5MHz, 2MHz, 6.1MHz, 10MHz, 20MHz, 40MHz, 62.6MHz, Bypass)
- Multiport capture memory
- Peak detector for fast peak voltage measurements
- Multiplexed, single-ended or differential inputs
- Input voltage range: 8mV to 8V peak
- Instrument initiated Move Client

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Introduction to the VHFAC Instrument : VHFAC Capture Instrument : Capture Instrument Characteristics

Capture Instrument Characteristics

The VHFAC capture instrument characteristics listed in the table provide a guide to its capabilities.

VHFAC Capture Instrument Characteristics

Characteristic	Value
High Resolution Sample Rate	0.003343 - 80Msps 14 bit resolution
High Speed Sample Rate	80 - 125Msps 12 bit resolution
High Voltage Single-Ended 10Kohm Peak Input Voltage (AC + DC)	(-20.48) - 20.474V DC - 9.5MHz frequency range
Differential AC Coupled 50 ohm Level Range, Peak differential	0.016 - 4.096V In 6 dB steps, DC - 61.5MHz frequency range
DC Baseline Removal, Single-ended DC Coupled 50ohm Input	2V
Sinewave Spurious (High Resolution Mode)	
Single-ended DC Coupled 50ohm Input	-60dBm, 100kHz to 10MHz frequency range
Differential DC Coupled 50ohm Input	-60dBm, 100kHz to 5MHz frequency range

Undersampling Mode Sampling Rate	3.434k - 20Msps 14 bit resolution
Peak Detector	
Frequency Range	0.1 - 100MHz 14 bit resolution
Peak Detector: Detection Pulse Width	2.0ns, typical
Peak Detector: Reset Pulse Width	1s, typical

For detailed specifications, see VHFAC Specifications.

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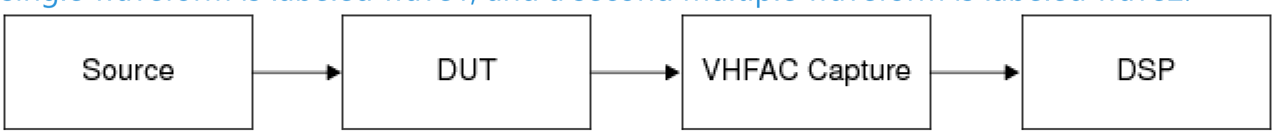
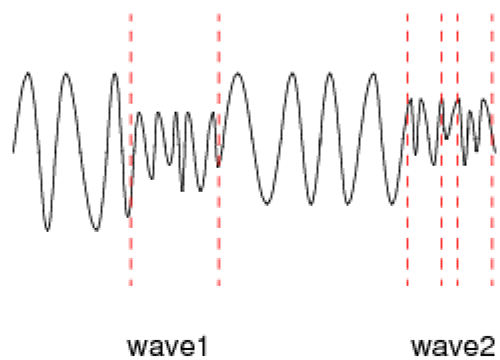
Introduction to the VHFAC Instrument : VHFAC Capture Instrument : Multicapture Capability

Multicapture Capability

The capture instrument can capture a single waveform or a block composed of multiple waveforms. When the capture is triggered, each capture is given a label. For example, in the figure below, one single waveform is labeled wave1, and a second multiple waveform is labeled wave2.





Capturing a Single Waveform and Multiple Waveform

Each waveform is identified as normal or block:

wave1 = normal, sz=1024

wave2 = block, sz=768

The waveforms, wave1 and wave2, go to the DSP and are processed separately to use less capture memory. Also the instrument keeps A_n running to maintain coherent capture. To manage all the waveforms captured with multicapture, a capture look-up table is used. In this example, wave1 and wave2 would be stored in two locations in the look-up table.

	Label	Size	Block Size	Mode
0	wave1	1024	-	Normal
	wave2	384	768	Block
1023				

Capture Look Up Table

Up to 1024 waves can be stored in the capture look up table per continuous capture executed.

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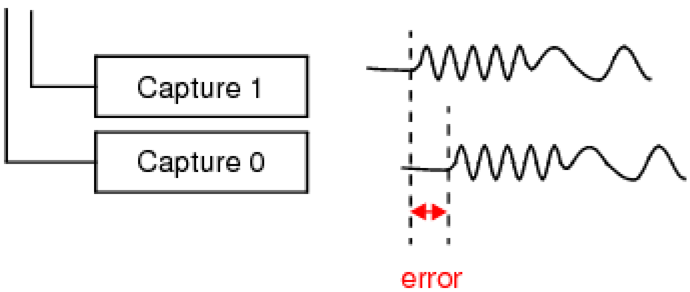
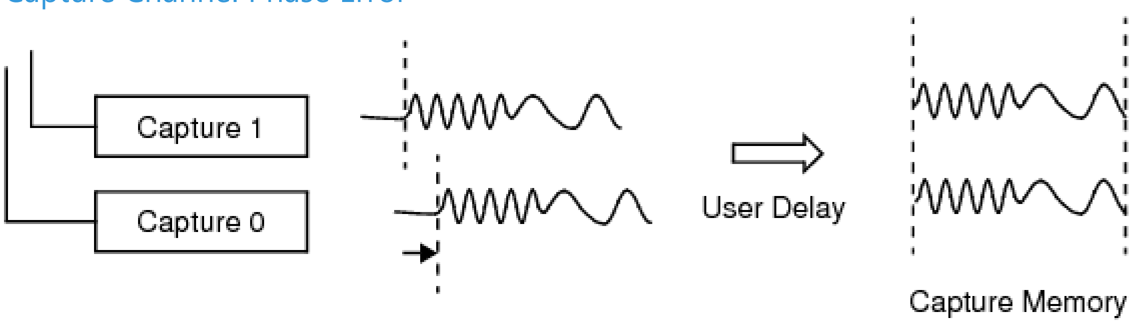
Introduction to the VHFAC Instrument : VHFAC Capture Instrument : User Delay Function

User Delay Function

As with the source instrument, the capture instrument can also use the DDS clock phase calibration to delay a capture to fine-tune its phase with respect to other capture channels.



Capture Channel Phase Error



Capture User Delay Function Allows Capture Phase Fine-Tuning

Here a programmed delay has been added to Capture 0.

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Introduction to the VHFAC Instrument : VHFAC Serial Bus

VHFAC Serial Bus

Each VHFAC board includes a serial bus which generates serial digital information useful for register setting and I^2C control.

Serial Bus Characteristics

The serial bus characteristics listed in the table provide a guide to its capabilities.

Digital Serial Bus for Register Setting and I^2C Control Characteristics

Characteristic	Value
Clock Channel	1
Drive/Receive	2
Vector Rate	0.0024 – 1.67MHz
Voltage Levels, No Load	(-4) –10V
Edge Placement Accuracy	30ns[1]
Pattern Memory	64kVectors

[1] Displacement between two channels programmed to the same point in time.

For detailed specifications, see VHFAC Specifications.

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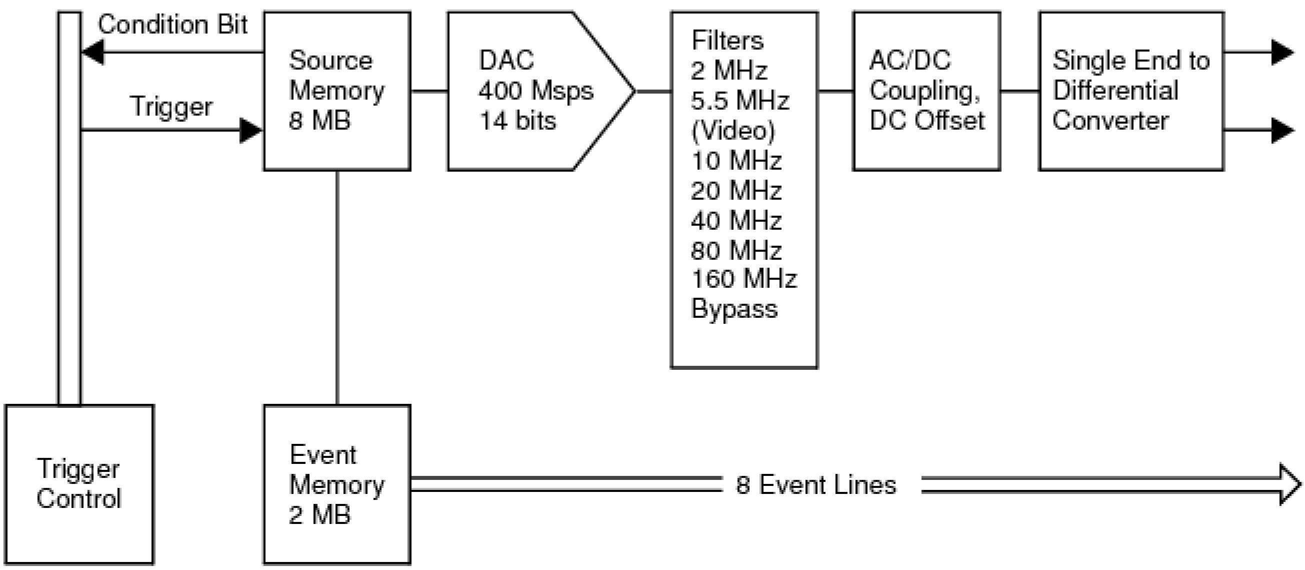
Introduction to the VHFAC Instrument : VHFAC Source Instrument

VHFAC Source Instrument

Each VHFAC board includes two independent source instruments. The following block diagram shows a single instrument.



VHFAC Source Instrument

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Introduction to the VHFAC Instrument : VHFAC Source Instrument : Features

Features

Key features of the source instrument are:

- High-speed, high-resolution DAC
- DC to 160MHz analog bandwidth
- Wide filter selection
- Harmonics: -70dBc @ 10MHz (-30 to 0dBm), -60dBc @ 40MHz (-30 to 0dBm)
- Spurs: -70dBm @ 10MHz (-30 to -10dBm), -55dBm @ 40MHz (-30 to -10dBm)
- Large waveform and event memories
- Event lines synced to Source for TV/VTR
- Multiplexed, single-ended or differential outputs

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Introduction to the VHFAC Instrument : VHFAC Source Instrument : Source Instrument Characteristics

Source Instrument Characteristics
The following list of VHFAC source characteristics provides a guide to its capabilities.

VHFAC Source Instrument Characteristics

Characteristic	Value
Sample Rate	0.003434 - 400Msps
Resolution	14 bits
Analog Bandwidth	DC - 160MHz
Peak Output Voltage: Single-Ended (AC + DC Baseline), DC Mode, Open Load	4V
Peak Output Voltage: Differential (H or L Output), AC Mode, 50ohm Load	3V
Programmable DC Baseline, 50ohm Load	2V
Waveform Memory	8Msamples
Event Lines	8
Event Memory	2Msamples
Sinewave Spurious DC - 10MHz, Level = -30 to -10dBm DC - 40MHz, Level = -30 to -10dBm	 -70dBm[1] -55dBm[2]

Noise, DC - 80MHz Range, 30kHz resolution BW	20Vms[3]

1 BW = 10kHz to 100MHz

2 BW = 10kHz to 150MHz

3 BW = 100Hz to 200MHz

For detailed specifications, see VHFAC Specifications.

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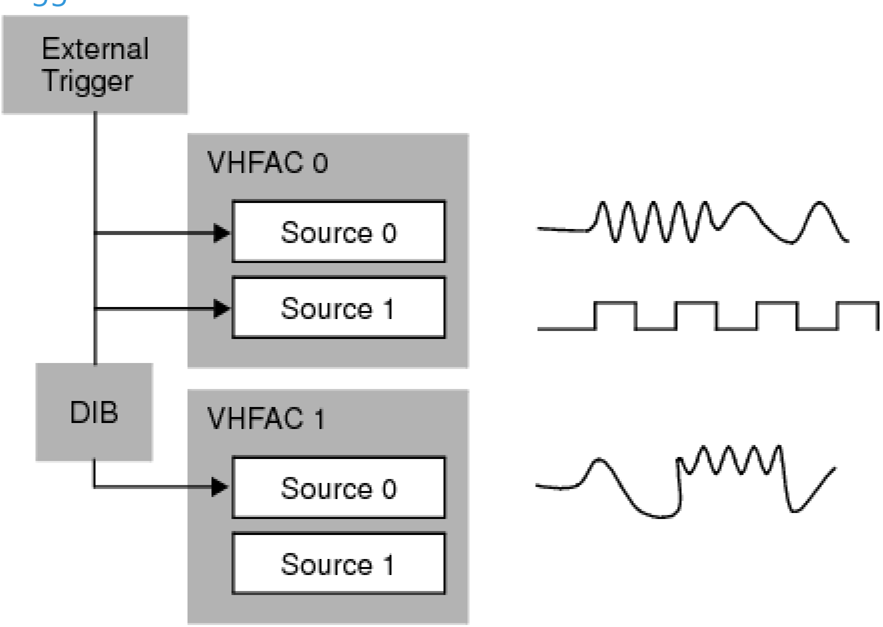
Introduction to the VHFAC Instrument : VHFAC Source Instrument : Source Synchronization

Source Synchronization

The VHFAC source hardware can be synchronized with the DUT by two methods: instructions in the pattern generator code or using trigger control.

Each VHFAC board contains a pattern generator (PatGen). Waveforms can be triggered through a digital pattern. Independent of the digital pattern, you can take advantage of the user delay function to fine-tune timing.

Waveforms can also be generated using a trigger. This trigger can be generated internally from a different VHFAC instrument or from some other instrument in the test system, or the trigger can come from the DUT over the DIB board. The analog path delay may be uncertain across boards or influenced by the DIB layout. To fine-tune the timing, the user delay function can be used with triggers also.



Synchronizing the Source Instrument Using an External Trigger

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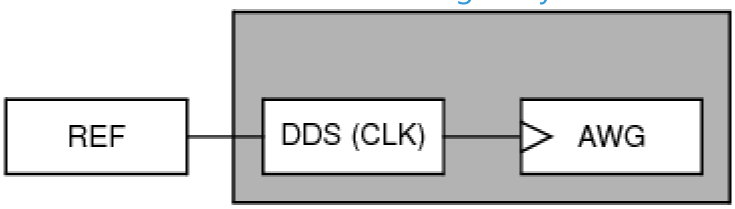
Introduction to the VHFAC Instrument : VHFAC Source Instrument : User Delay Function

User Delay Function

VHFAC instruments use direct digital synthesis to create a sampling frequency signal.

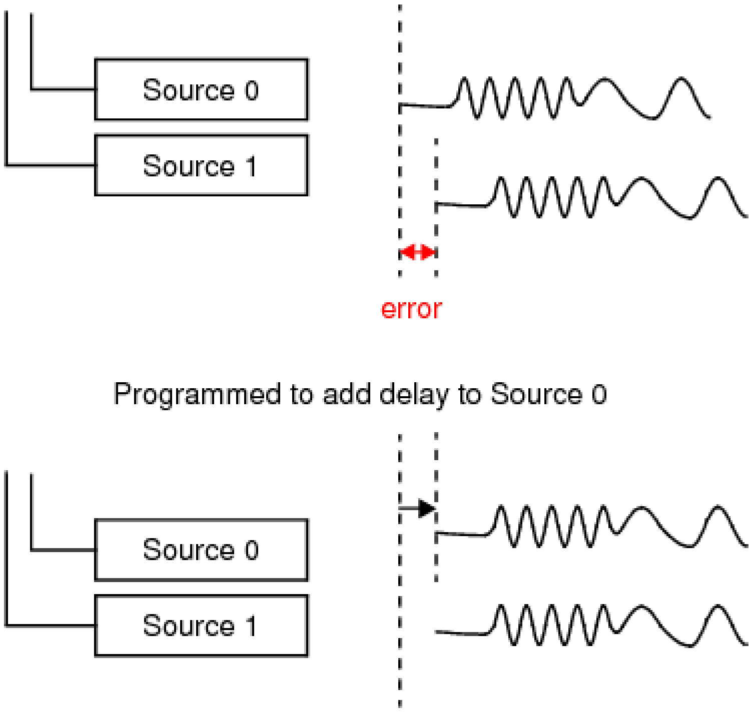


VHFAC Direct Digital Synthesis

This synthesizer can achieve femtosecond resolution. However, period accuracy is more limited. This is reflected in the sampling frequency edge resolution determined by the amount of jitter on the test system clock.

Users typically want to adjust waveform timing to find the precise point at which a DUT is likely to fail. The user delay function makes this possible.

The User Delay function is accomplished using DDS Clock Phase Calibration. User delay requires exact A_n placement, precise microcode tossing, and a Tunable A_n clock pipeline. With these capabilities, you can shift waveforms as shown in the figure.



User Delay for the Source Instrument

In this figure, delay is added to source 0. You can add delay to either source 0 or source 1. Adding a delay makes it easier to program a waveform's phase. Test systems without user delay would require padding waveforms to fine-tune the source channel's phase. Here you can include a delay without modifying the waveform definition. You can use this feature with both synchronization methods.

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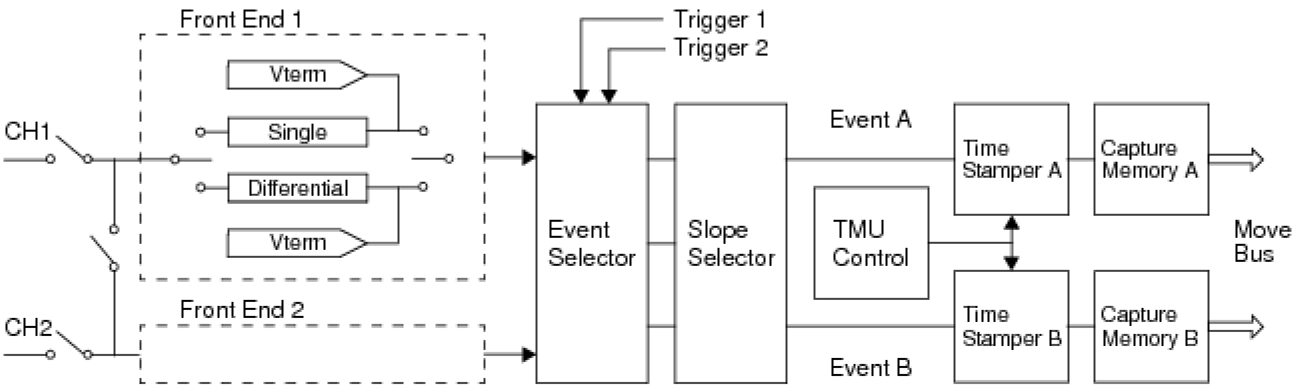
Introduction to the VHFAC Instrument : VHFAC Time Measurement Unit (TMU)

VHFAC Time Measurement Unit (TMU)

Each VHFAC board has a Time Measurement Unit (TMU) option. The TMU is illustrated in the following block diagram.



VHFAC TMU Instrument

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Introduction to the VHFAC Instrument : VHFAC Time Measurement Unit (TMU) : Features

Features

The features of the TMU are:

- Period, Frequency, Duty Cycle, Pulse Width, Rise/Fall Time, Propagation Delay, Event Capture, Jitter Measurements
- Resolution: low range 2.44ps, high range 5ns
- 80ns minimum capture cycle
- 512k Sample memory depth

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Introduction to the VHFAC Instrument : VHFAC Time Measurement Unit (TMU) : TMU Characteristics

TMU Characteristics

The VHFAC TMU characteristics listed in the table provide a guide to its capabilities.

VHFAC TMU with Dual Time Stampers Characteristics

Characteristic	Value
Resolution	
Time Range: Low	2.44ps nominal
Time Range: High	5ns nominal
Time Base	200MHz
Time Base Error, Synchronous Mode	1ppm
Time Between Stamps, Time Range: Low	80ns
Time Stamp Jitter	75ps rms maximum
Time Stamper Accuracy	(Resolution + TBE*Time + Jitter)
Memory Depth	512kSamples
Input Voltage, Single-ended, 50ohms	5kV
Threshold Range, Single-ended, 50ohms	5V
Hysteresis Range	1 – 40mV typical
Hysteresis Resolution	0.5mV typical

For detailed specifications, see VHFAC Specifications.

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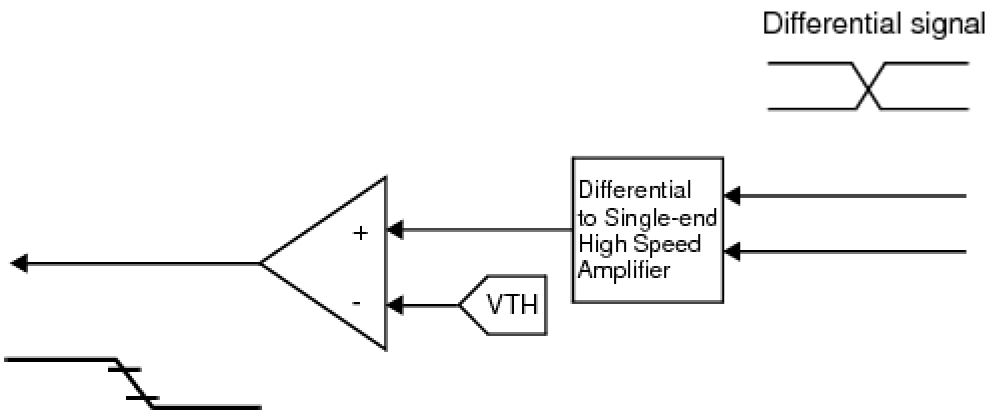
Introduction to the VHFAC Instrument : VHFAC Time Measurement Unit (TMU) : Rise/Fall Time Measurement on Differential Signals

Rise/Fall Time Measurement on Differential Signals

The TMU can measure ordinary rise/fall times and differential rise/fall times. As shown in this figure, the TMU uses a differential to single-end amplifier to make differential rise/fall time measurements.



Measuring Rise and Fall Times Using the TMU

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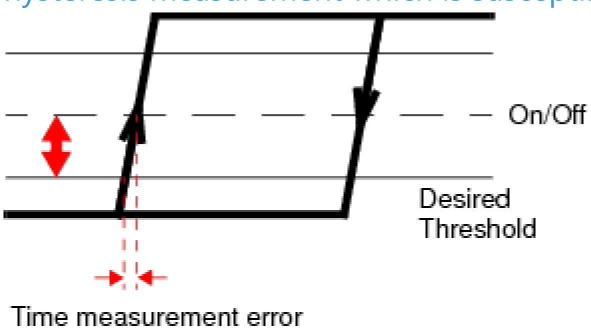
Introduction to the VHFAC Instrument : VHFAC Time Measurement Unit (TMU) : Programmable Hysteresis

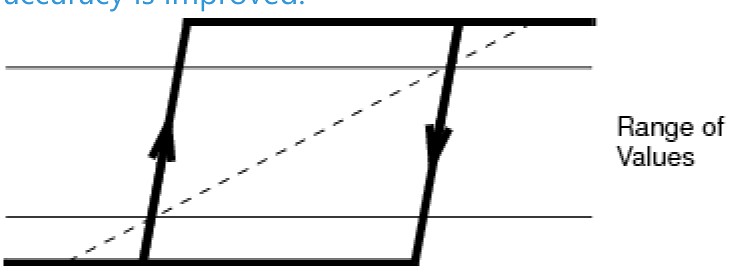
Programmable Hysteresis

The VHFAC TMU can provide programmable hysteresis. Conventional techniques only allow on/off hysteresis measurement which is susceptible to noise.



Hysteresis Error Produces Time Measurement Error

With the TMU's programmable hysteresis feature, stable measurements can be made and threshold accuracy is improved.



TMU Programmable Hysteresis Helps Make Stable Measurements

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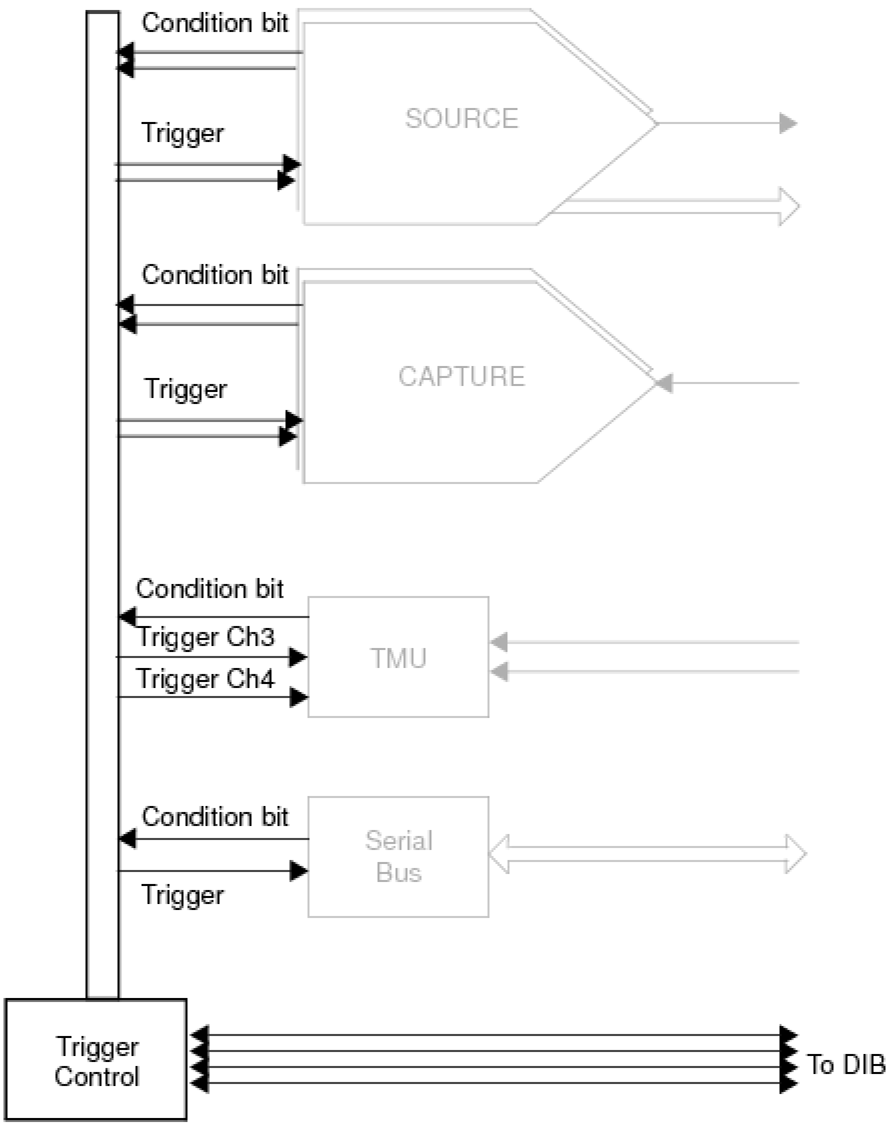
Introduction to the VHFAC Instrument : VHFAC Trigger Control

VHFAC Trigger Control

The VHFAC Trigger Control supports the operation of the instruments on the board.



VHFAC Trigger Control

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Introduction to the VHFAC Instrument : VHFAC Trigger Control : Features

Features

The features of Trigger control are:

- Four trigger I/O lines to DIB
- Access to all instruments
- Ability for instruments to trigger other instruments and the DIB

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Introduction to the VHFAC Instrument : VHFAC Trigger Control : Characteristics of Trigger Control

Characteristics of Trigger Control

The Trigger Control characteristics listed in the table provide a guide to its capabilities.

Trigger Control To and From Instruments and the DUT

Characteristic	Value
Trigger I/O Channels	4
DC Input and Output Levels:	
VIL	(-0.5) – 0.8V typical
VIH	2.0 – 3.6V typical
VOL	0.4V typical maximum
VOH	2.4V typical minimum
IOL	24mA typical minimum
IOH	-24mA typical minimum
Input Leakage Current	10A typical
Propagation Delay	10ns typical
Minimum Input Pulse Width	20ns typical

For detailed specifications, see VHFAC Specifications.

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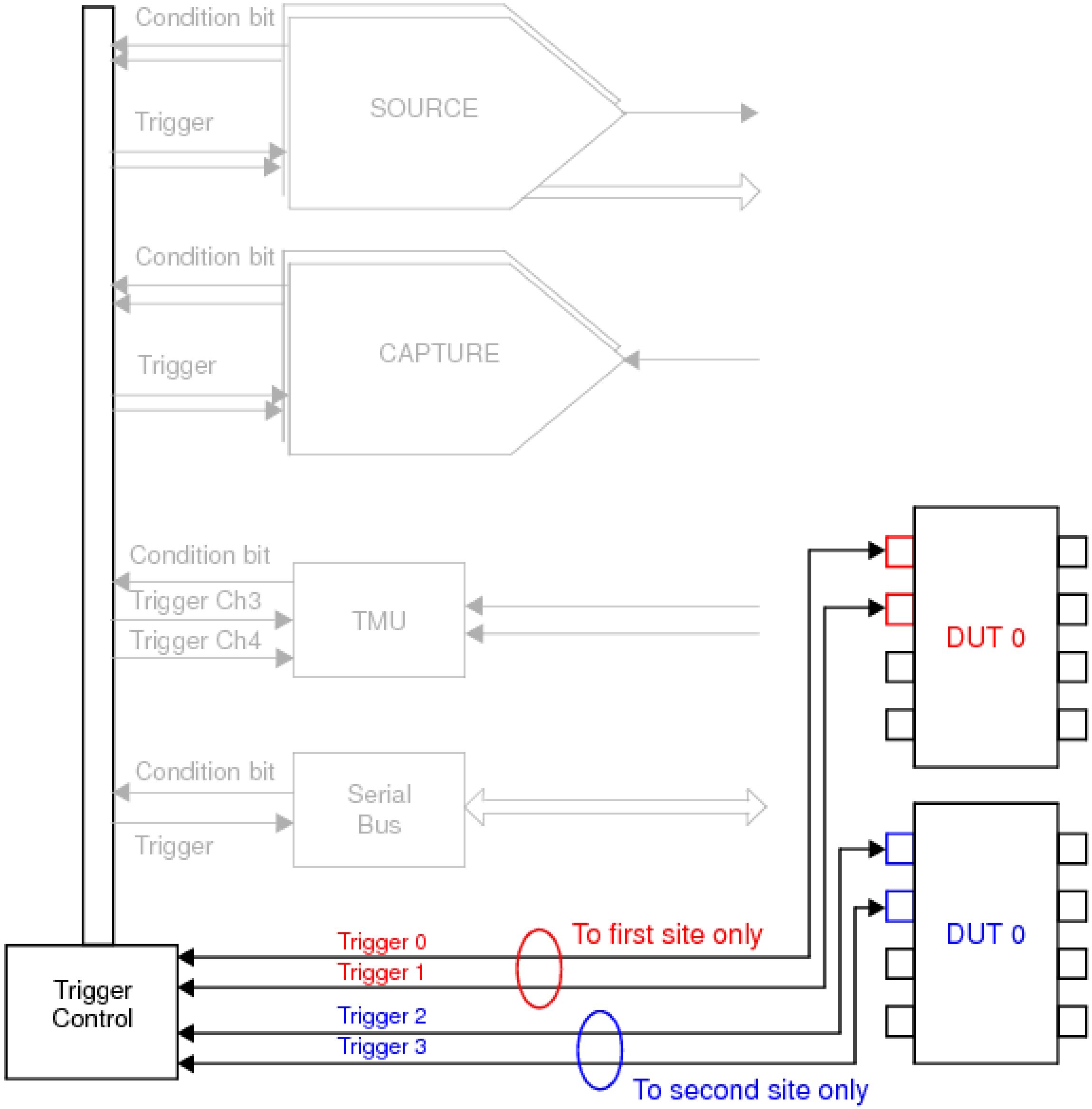
Introduction to the VHFAC Instrument : VHFAC Trigger Control : Trigger Control for Multisite Assignment

Trigger Control for Multisite Assignment
Trigger control can be set up so that different DUTs can trigger or be triggered by the same VHFAC board. This increases the multisite test capability of the UltraFLEX test system.



Trigger Control with Two DUTs

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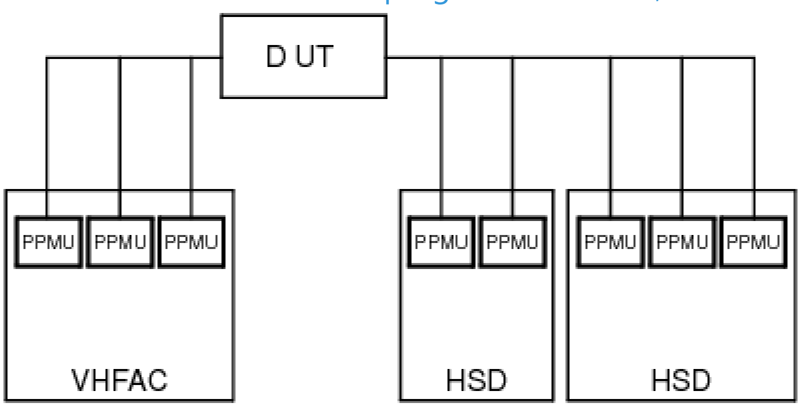
Introduction to the VHFAC Instrument : VHFAC PPMU

VHFAC PPMU

Each VHFAC includes 23 pin parametric measurement units (PPMUs). These PPMUs are functionally equivalent to other PPMUs implemented on other UltraFLEX instruments. The PPMUs allow you to make parameter measurements with multiple resources spread across different board types for greater flexibility.

For information on how to program the PPMU, see About the PPMU.



UltraFLEX Test System Distributed PPMUs

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Introduction to the VHFAC Instrument : VHFAC Software Licensing Requirements

VHFAC Software Licensing Requirements

IG-XL enforces the following node-locked licenses for VHFAC Instruments:

VHFAC Licenses

License Type	License Name	Enabling Points
Source Channels	VHFACSource	In channels per tester: 48, 46, 44, 42, 40, 38, 36, 34, 32, 30, 28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 2, 1, 0
Source Sampling Rate	VHFACSourceSamplingRate	In Msps: 400, 100
Capture Channels	VHFACCapture	In channels per tester: 48, 46, 44, 42, 40, 38, 36, 34, 32, 30, 28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 2, 1, 0
Capture Sampling Rate	VHFACCaptureSamplingRate	In Msps: 125, 80

Online, IG-XL obtains these licenses from the license file. Offline, the simulated configuration file defines these counted licenses under the license enablers section.

In an offline simulated or current configuration file, these licenses appear as follows:

```
<LICENSE_ENABLERS>
#Off-line enabling levels
#FEATURE          ENABLED LEVEL   LICENSE COUNT   LEGAL ENABLING POINTS
VHFACCapture        24             N/A            48,46,44,42,40,38,36,34,
```

			32,30,28,26,24,22,20,18, 16,14,12,10,8,6,4,2,1,0
VHFACCaptureSamplingRate	125	N/A	125,80
VHFACSource	24	N/A	48,46,44,42,40,38,36,34, 32,30,28,26,24,22,20,18, 16,14,12,10,8,6,4,2,1,0
VHFACSourceSamplingRate	400	N/A	400,100

</LICENSE_ENABLERS>

The Serial Bus, Time Measurement, and Trigger Control instruments do not require software licenses.

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Introduction to the VHFAC Instrument : Using VHFAC in Concurrent Testing

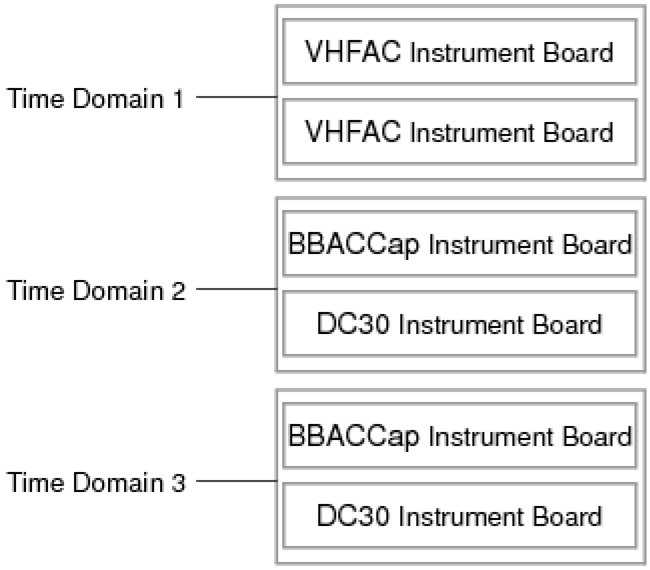
Using VHFAC in Concurrent Testing

To support concurrent testing, you can assign VHFAC channels to a time domain. This means that VHFAC instruments can be part of a concurrent pattern burst. However, only one pattern in the concurrent burst can contain VHFAC instruments in it at one time. This limitation means that VHFAC channels cannot be in more that one time domain for each concurrent burst.

Concurrent features for VHFAC include:

- A VHFAC board can be configured for one time domain
- Multiple VHFAC instruments can be grouped into a common time domain
- VHFAC is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with two VHFAC instruments assigned to its one time domain and other UltraFLEX instruments assigned to other time domains.



VHFAC Concurrent Testing Configuration Example

For more information, see About Concurrent Test.

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